Engineering Department
National Institute of Technology Kurukshetra

Subject: Power Electronics-1
Mid. Sem. - II
Note: Attempt all questions.

Dated: 09/04/25
Time: 50 Minutes
Max. Marks: 15

Q1. Design and explain the operation of three-phase semi- converter with RL load. Draw the output waveforms w.r.t. input supply and derive the expression for RMS output voltage at $\alpha<60^0$ and $\alpha\geq60^0$.

Q2. A battery is charged from 1-ϕ supply mains of 230V, 50 Hz through a 1-ϕ fully controlled bridge converter. The battery emf is 180V and its internal resistance is 0.5Ω. An inductor connected in series with the battery renders the charging current of 20A ripple-free. Compute the firing angle delay and the supply power factor.

Q3. Demonstrate the working operation of the 1-ϕ full wave half controlled converter with RLE load for α firing angle with output voltage, current and thyristor voltage waveform w.r.t. input supply in discontinuous conduction mode. Also, derive the expression for average output voltage.